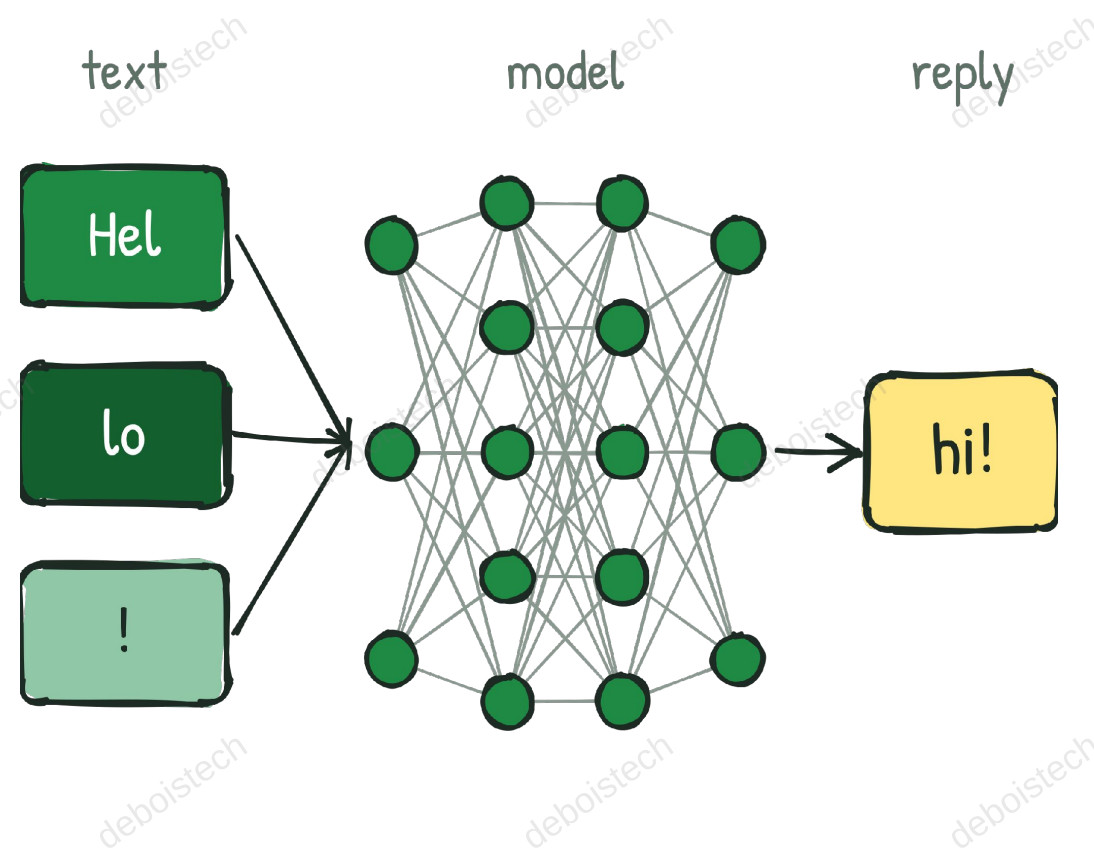


LLM Foundations

Tokens · Context · Inference



DAY 1 · ASTRA · TRACK 1, WEEK 1

The hook

Today we answer it mechanically, not vaguely.

Why does ChatGPT
forget what I said
20 messages ago?

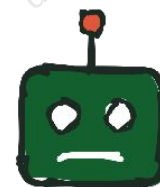


My name is Sam.

... 20 messages later ...



What's my name?



Sorry, I don't know



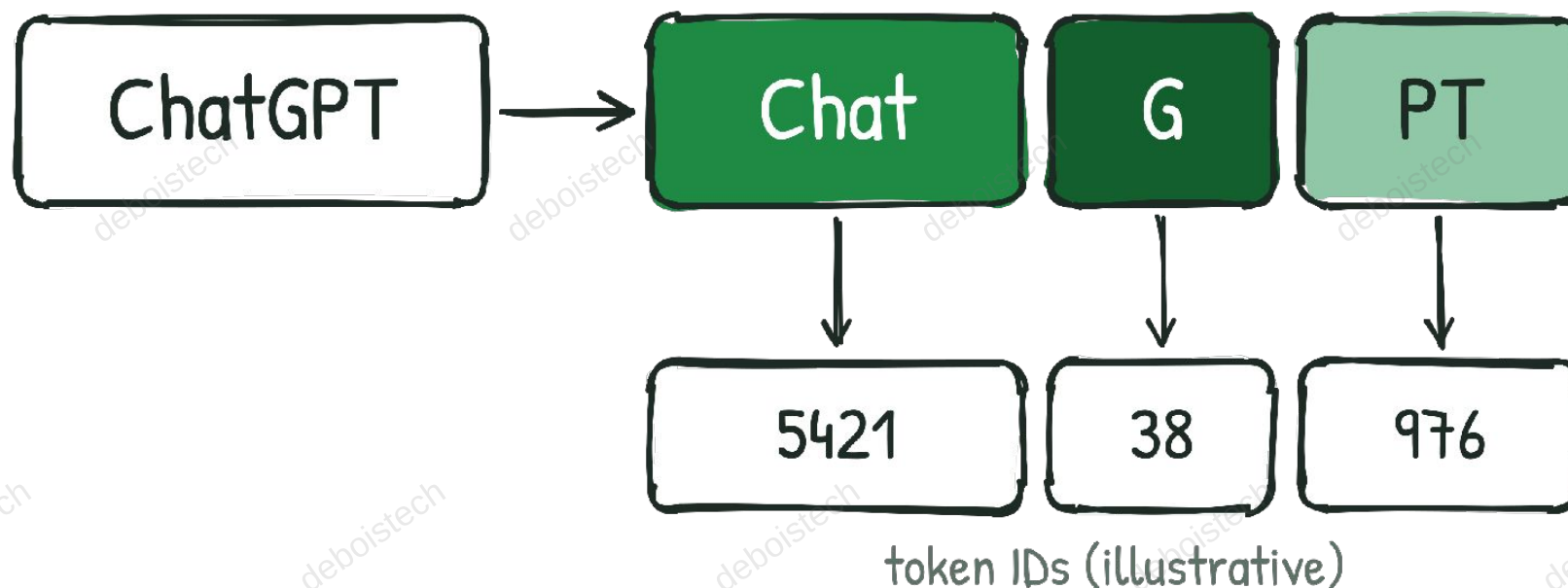
What is a token?

Models read tokens, not text.

~~words~~

~~characters~~

subwords

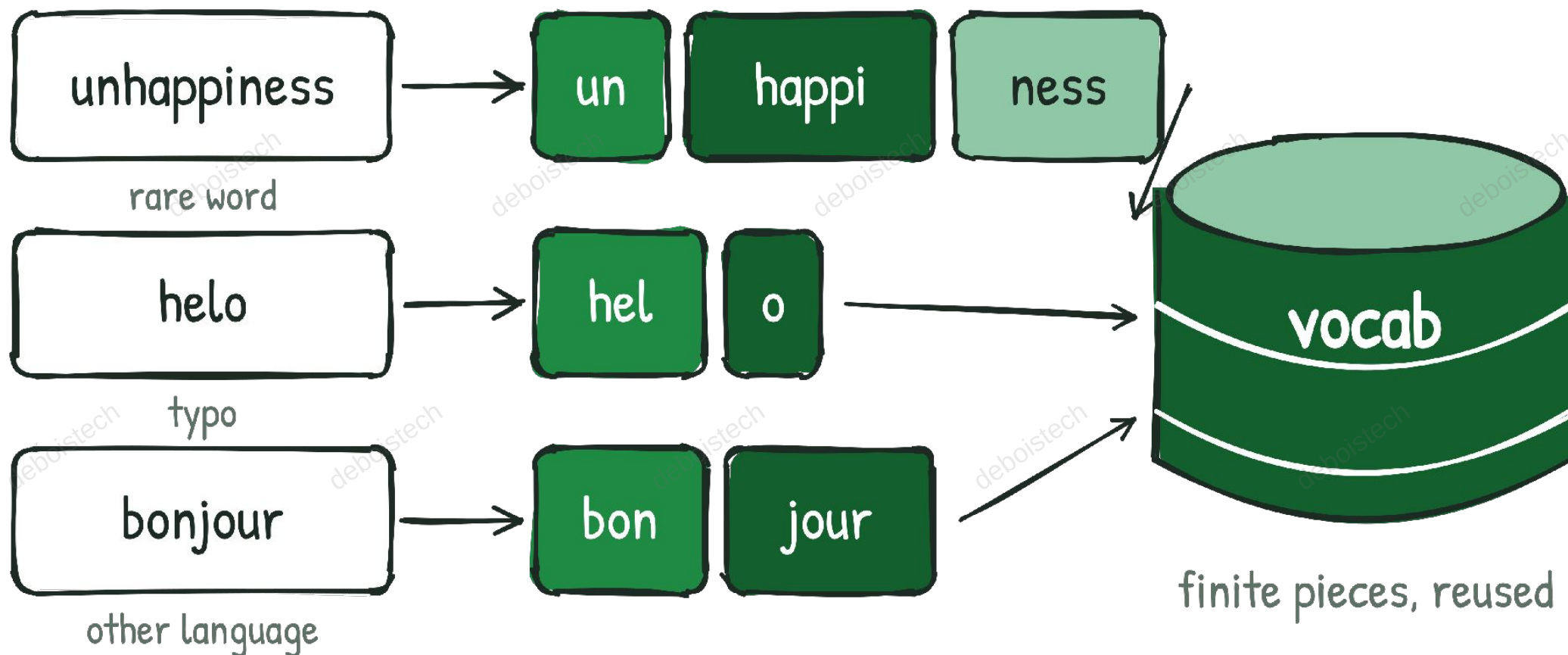


text > tokens > numbers

3 tokens

Why tokenization exists

One finite vocabulary, endless text.



BPE · Byte-Pair Encoding

Merge the most frequent pair. Repeat.

start: characters

l o w e s t

e + s is common

l o w es t

es + t is common

l o w est

l + o + w is common

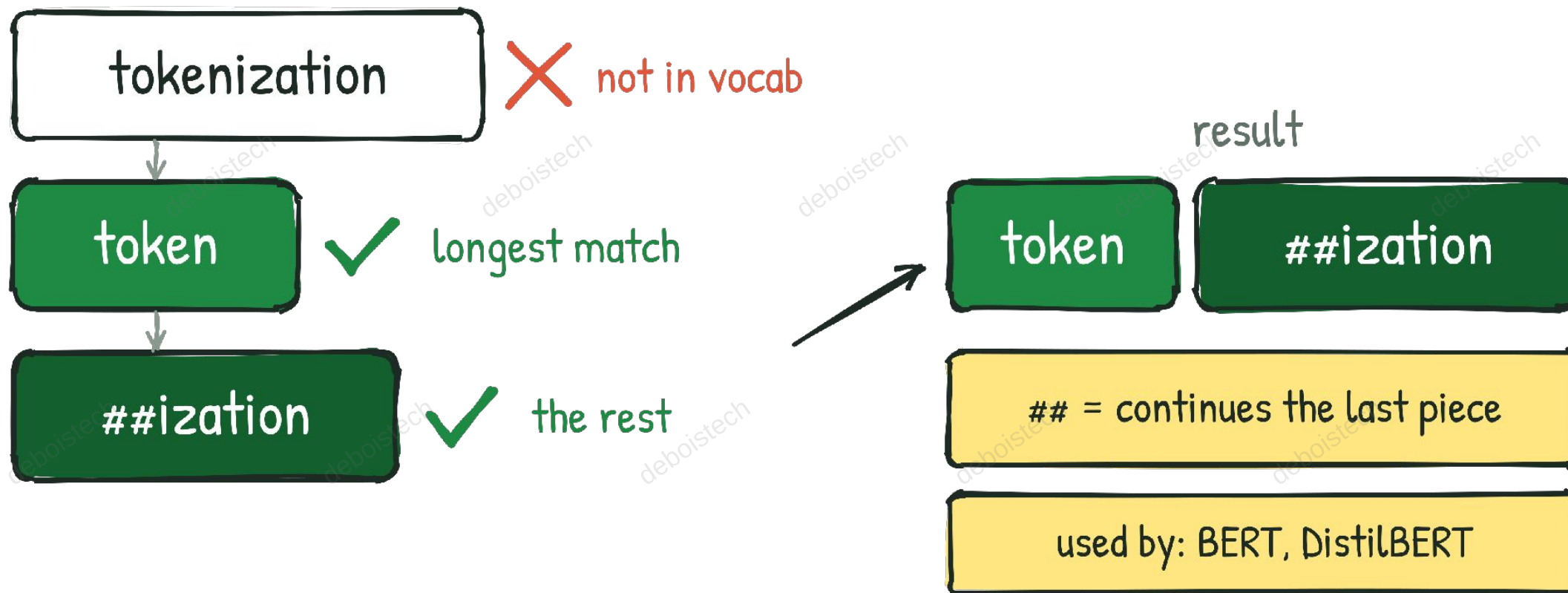
low est

repeat until the vocabulary is full

used by
GPT-2 / 3
RoBERTa
Llama 3

WordPiece

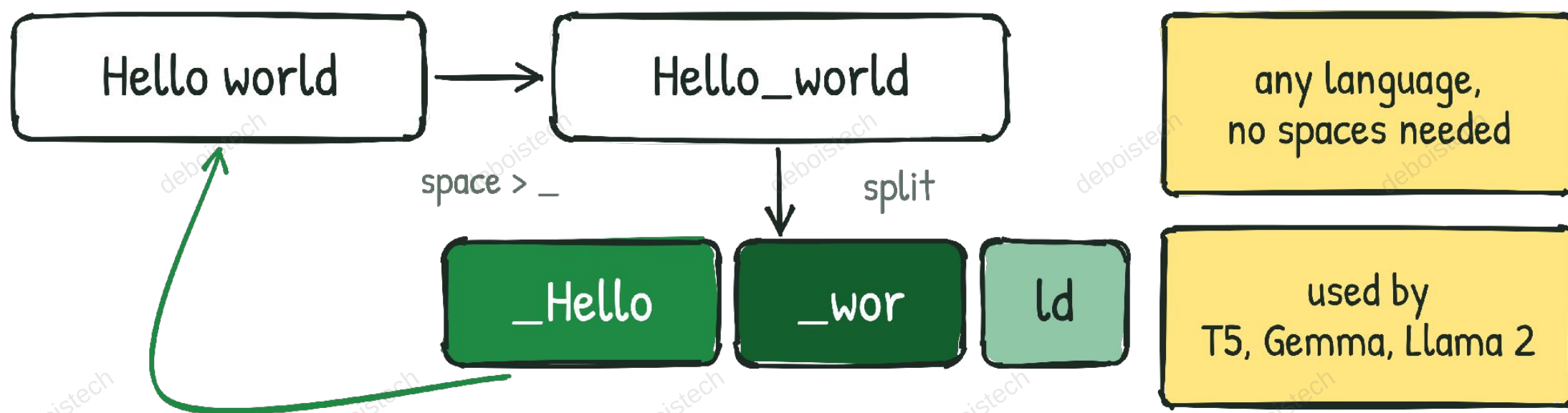
Greedy longest match. ## marks a continuation.



training merges the pair with the best $\text{count}(ab) / (\text{count}(a) \times \text{count}(b))$

SentencePiece

Works on raw text. A space is just a symbol.

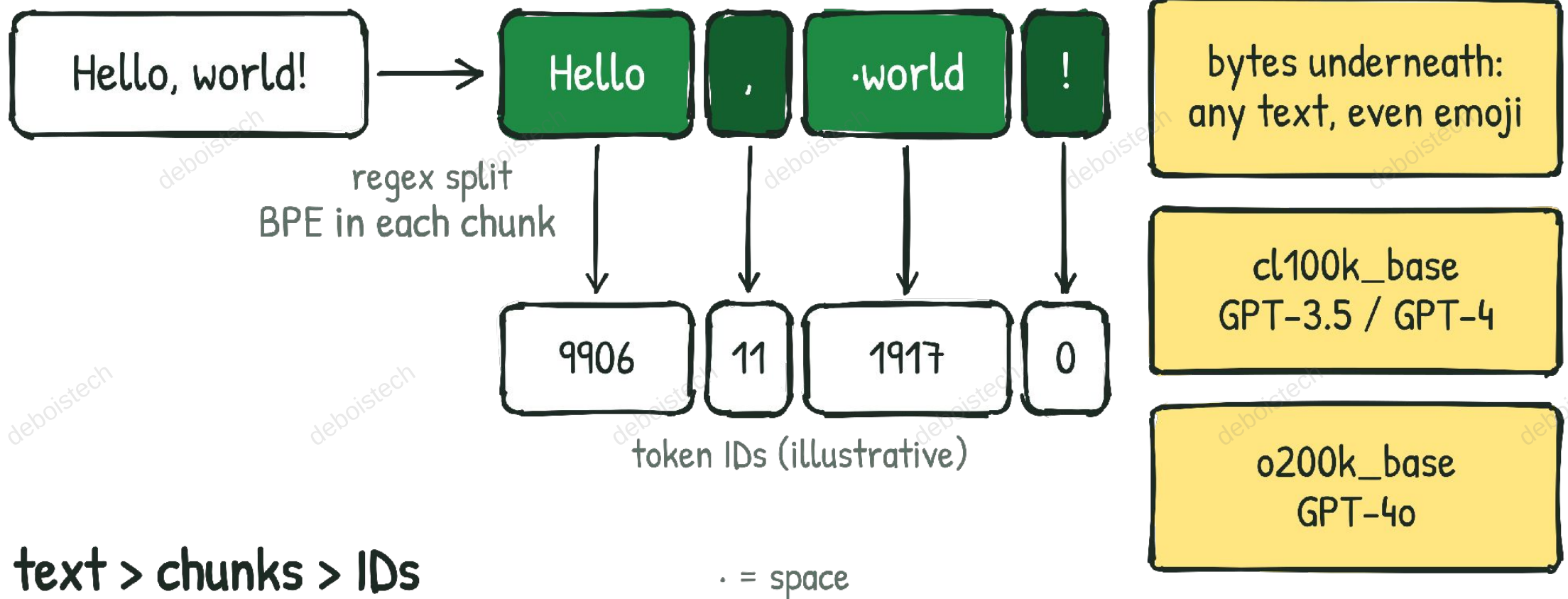


decode: join, _ to space
lossless round trip

engine inside: BPE or Unigram

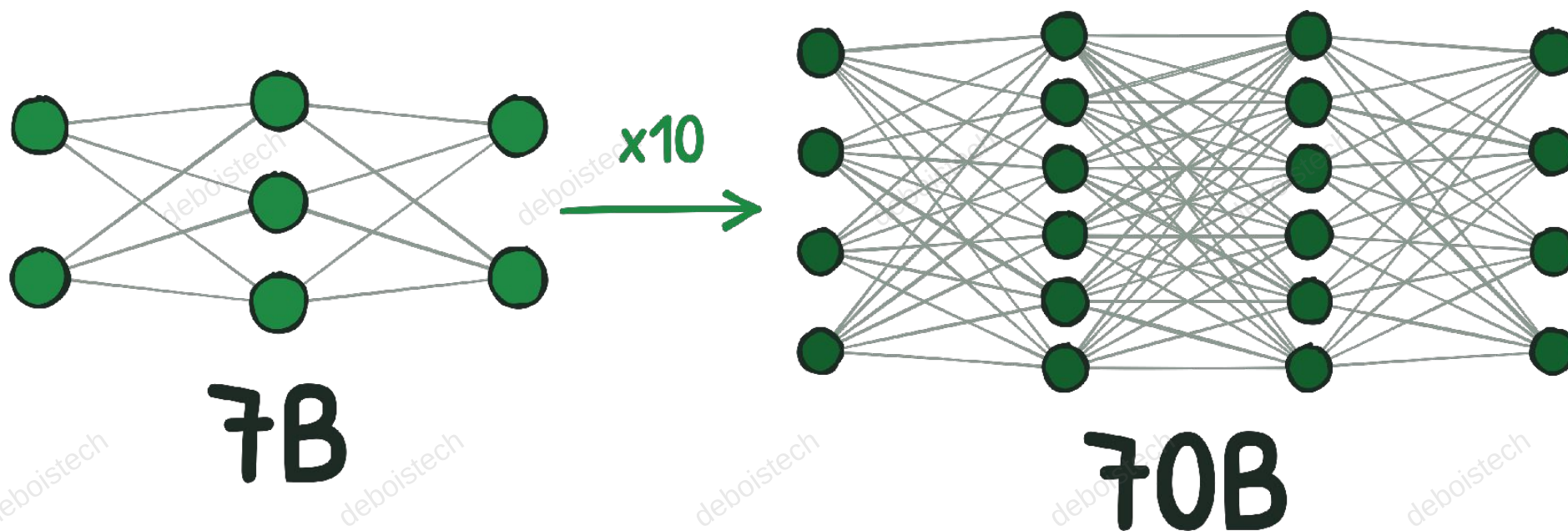
tiktoken

OpenAI's fast byte-level BPE.



Parameters = model size

Billions of tunable numbers.

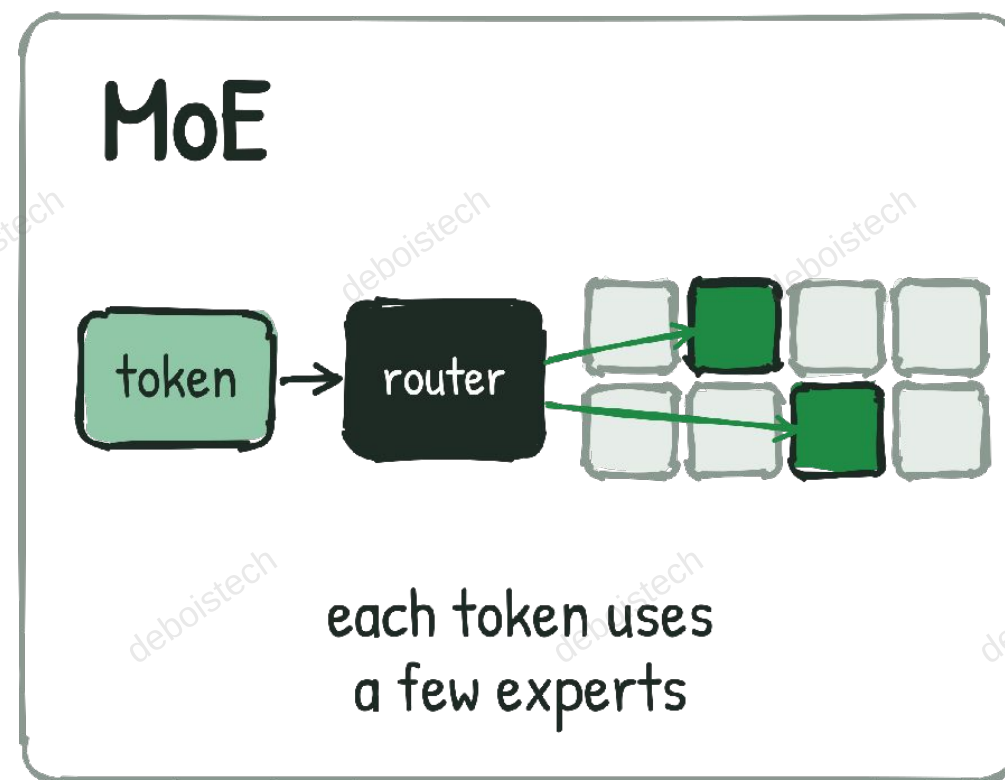
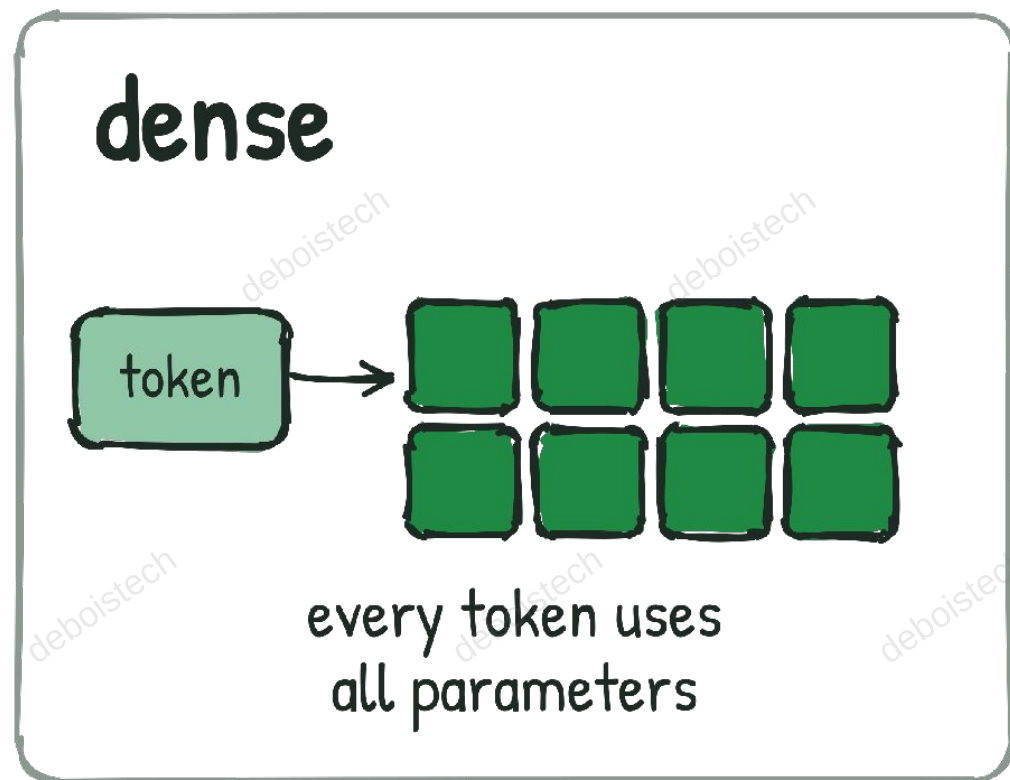


bigger is not always better

● cost ⌚ speed ✨ skill

Dense vs. MoE

Mixture of Experts: only some experts run per token.

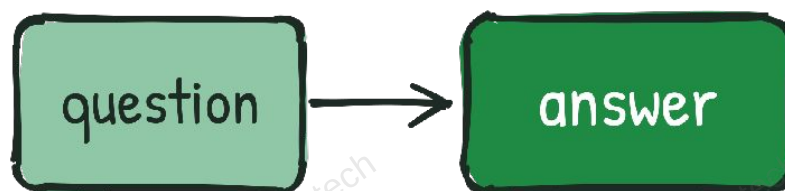


total params $\neq$ active params

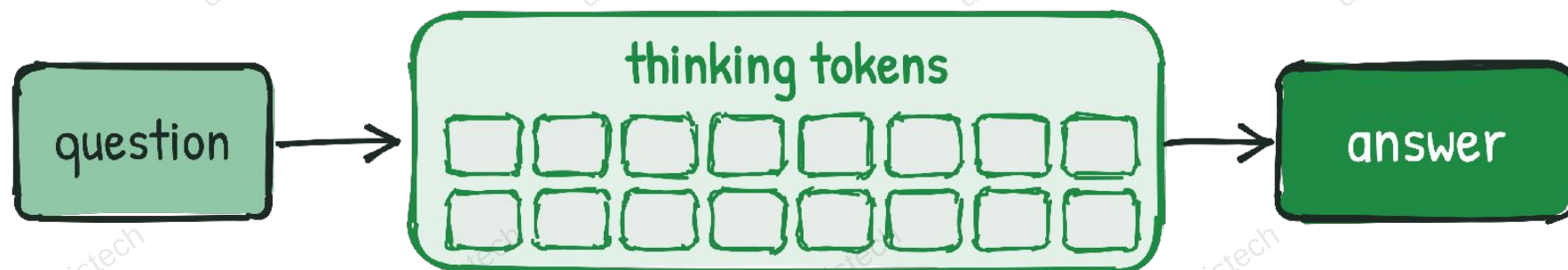
Reasoning models

They think in tokens before they answer.

standard



reasoning



more tokens



slower answer



more cost

Types of models

Pick the right kind for the job.

●●● | **Base**

predicts the
next token



Instruct

tuned to follow
your instructions



Multimodal

text + images
+ audio



Small

distilled, runs on
a laptop or phone

[0.2] **Embedding**

text in,
numbers out

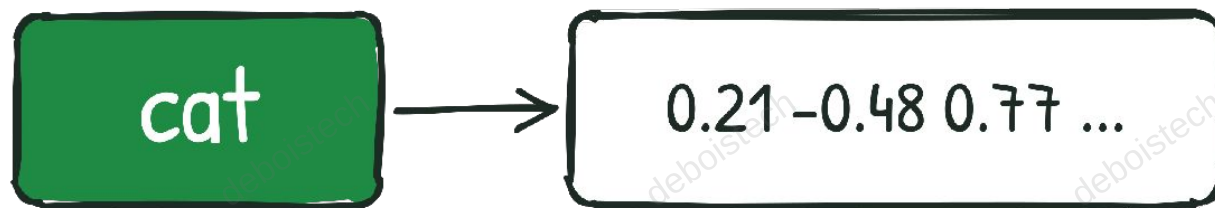


Open

weights you can
download and host

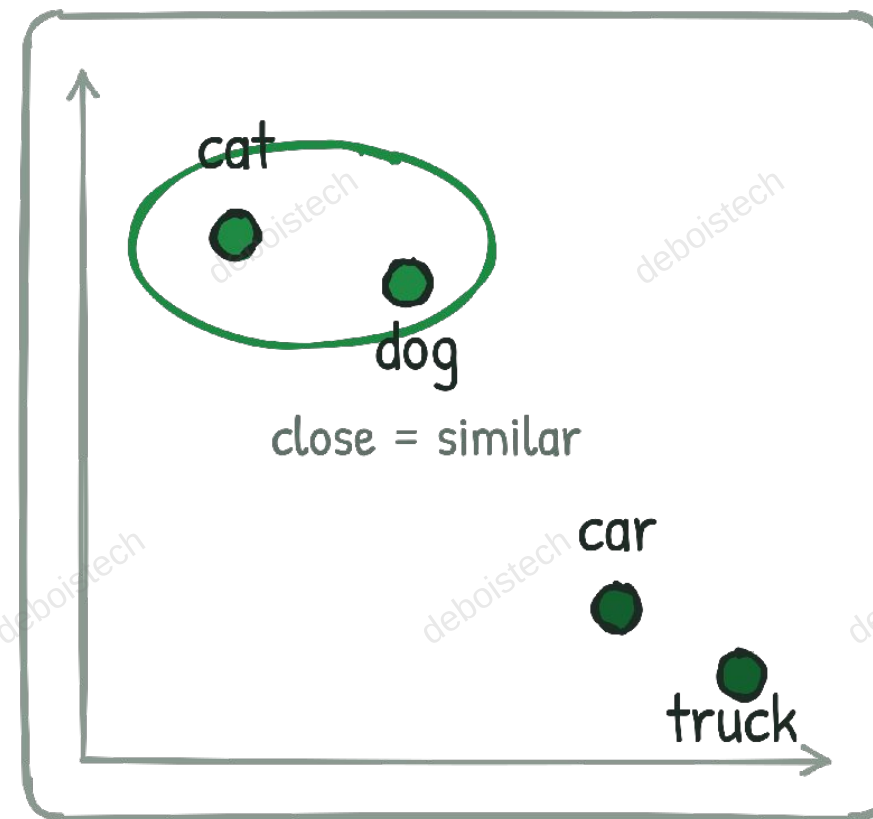
Embeddings

Meaning, turned into numbers.



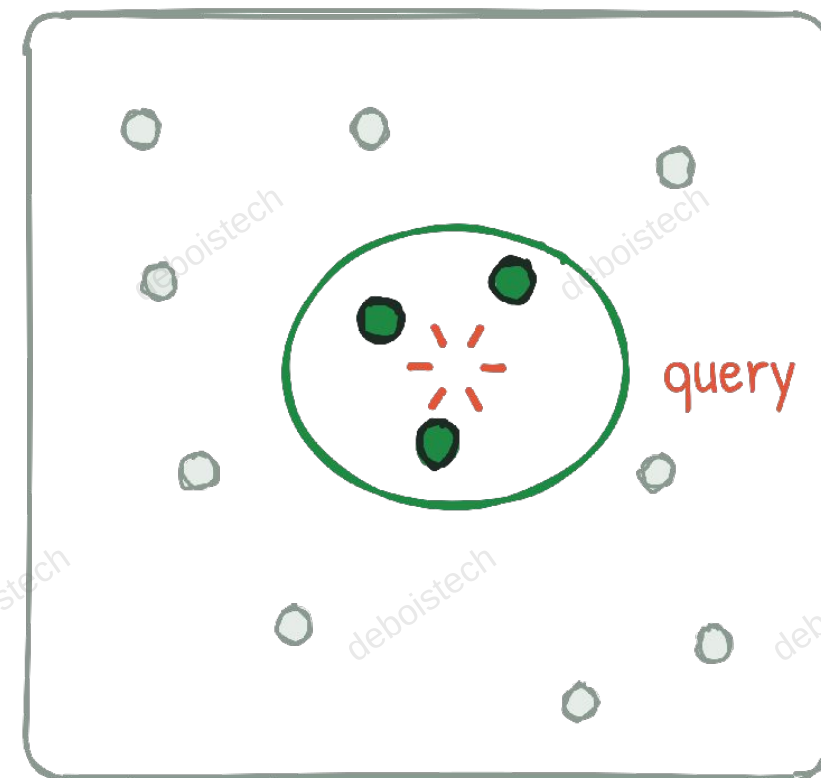
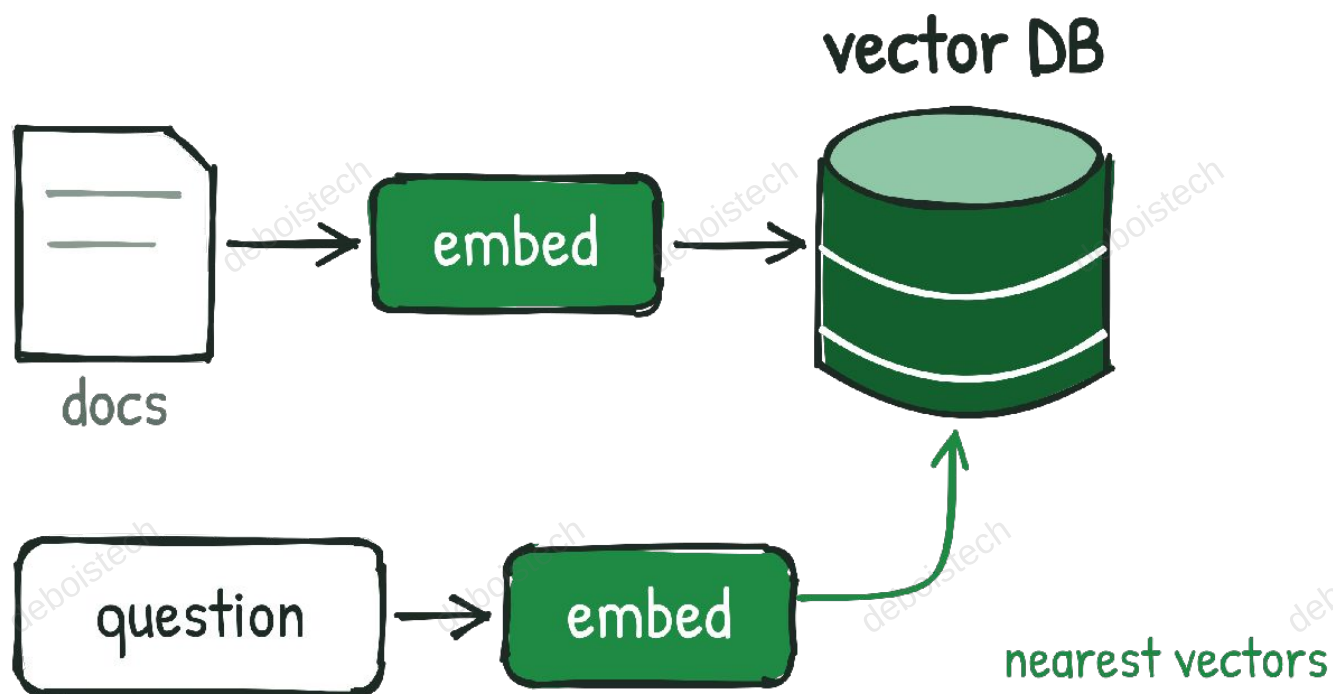
meaning > numbers

Day 4: RAG goes deep



Vector databases

Search by meaning, not by keywords.

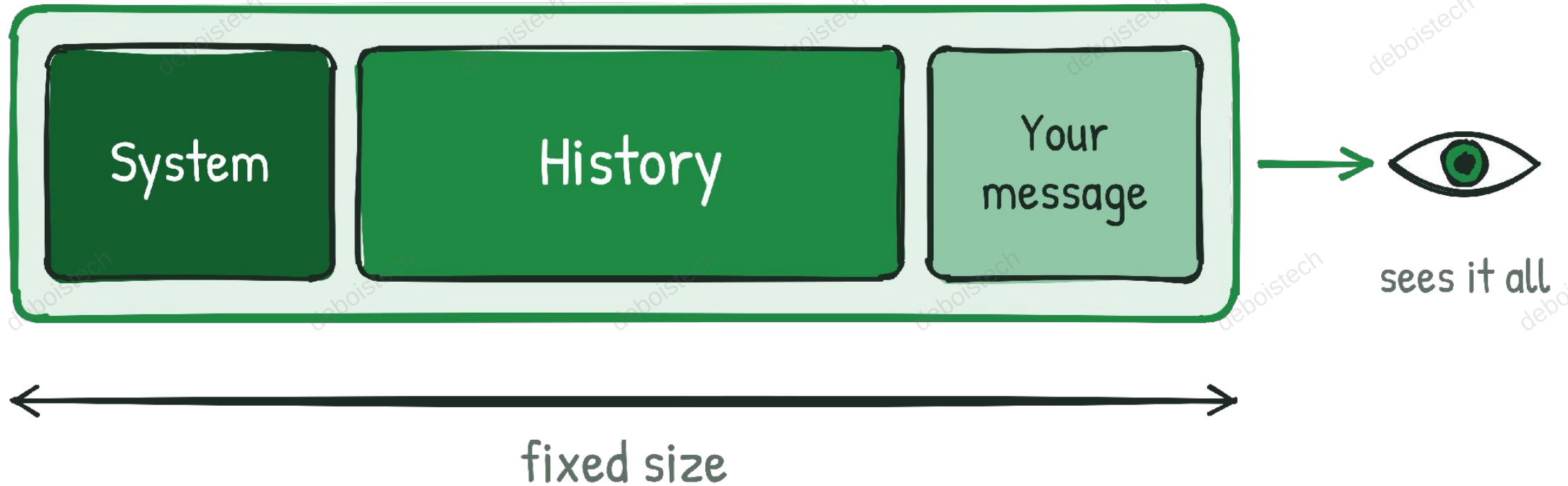


Chroma · pgvector · Pinecone · Qdrant · FAISS

Context window

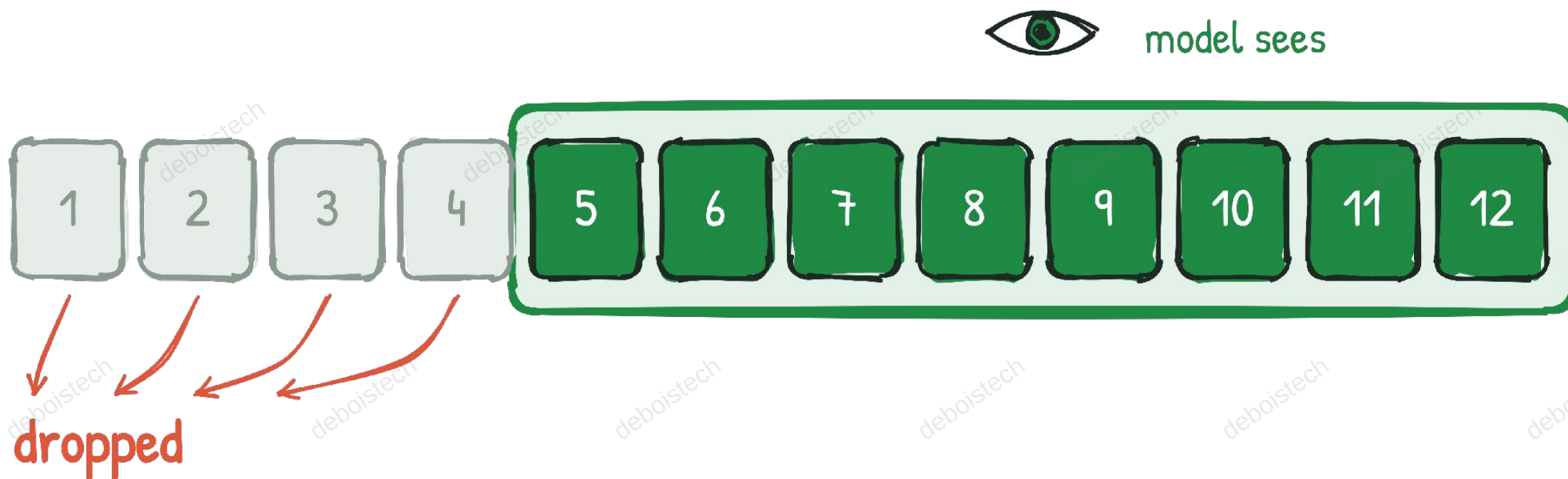
A fixed-size input buffer, not memory.

context window



Why long chats forget

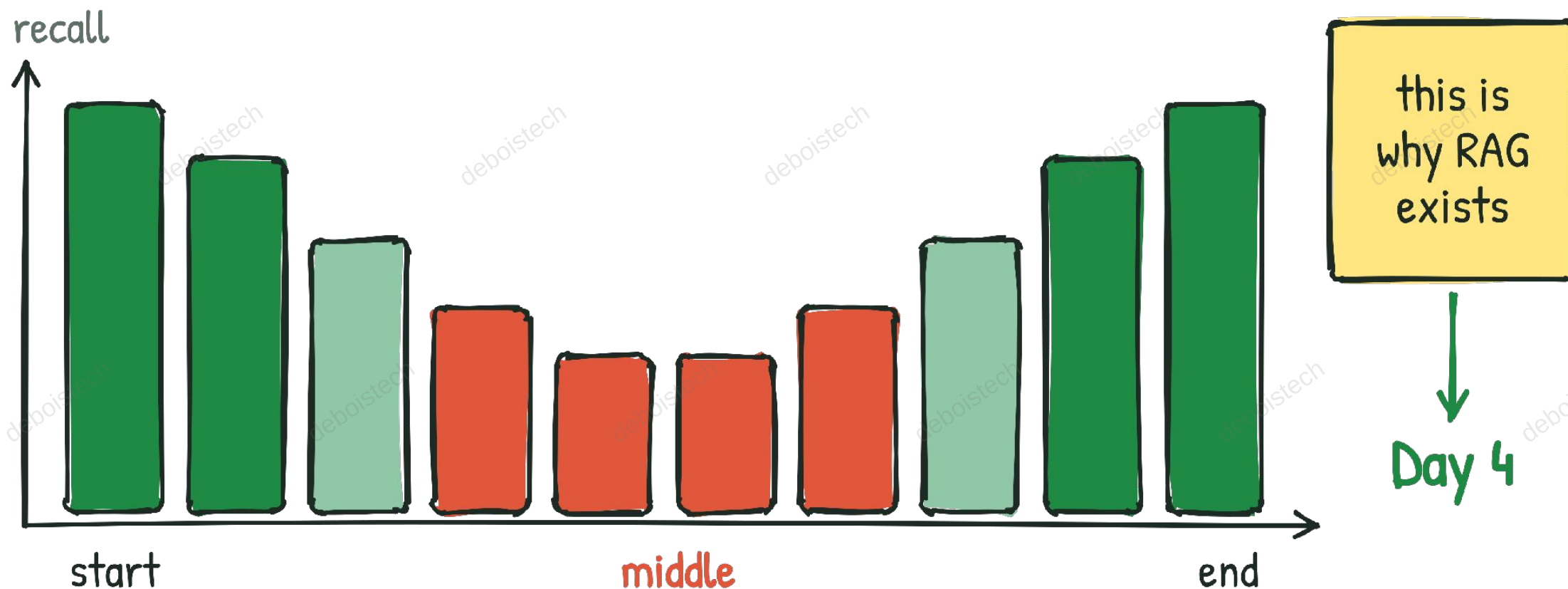
The hook, answered.



Not forgetting. Can't see.

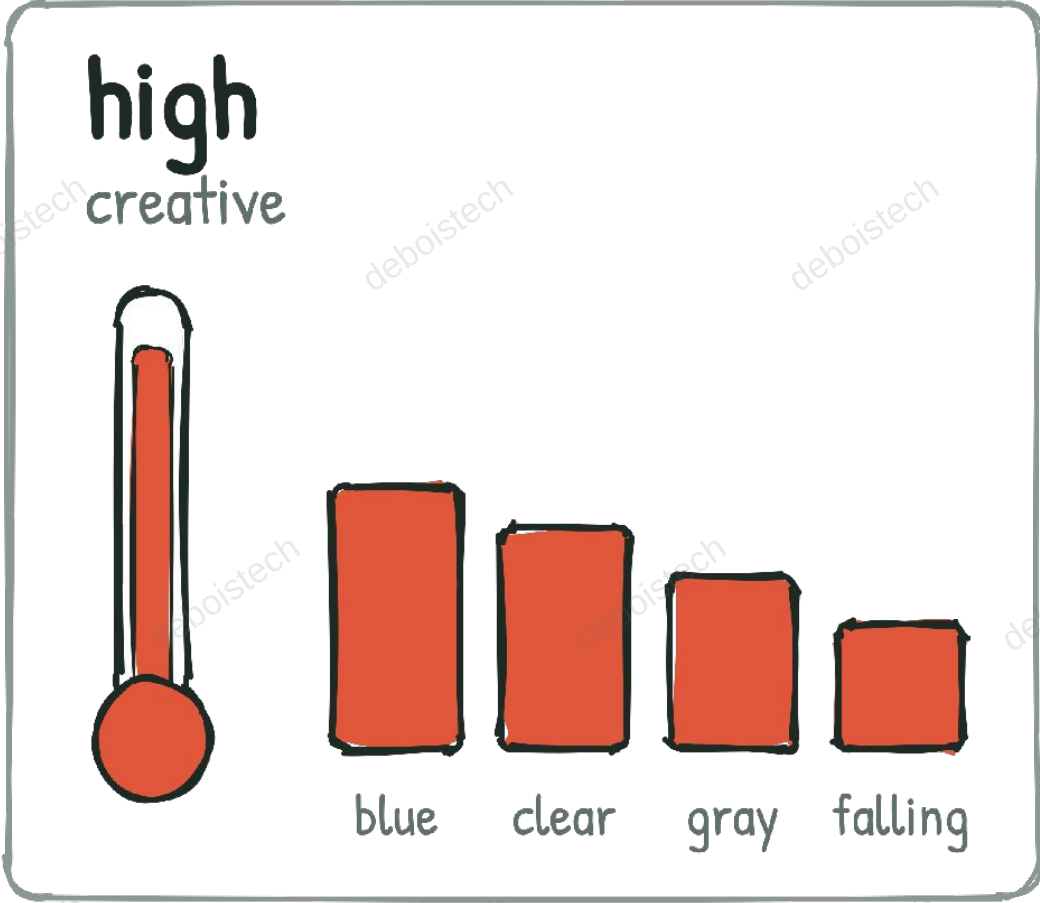
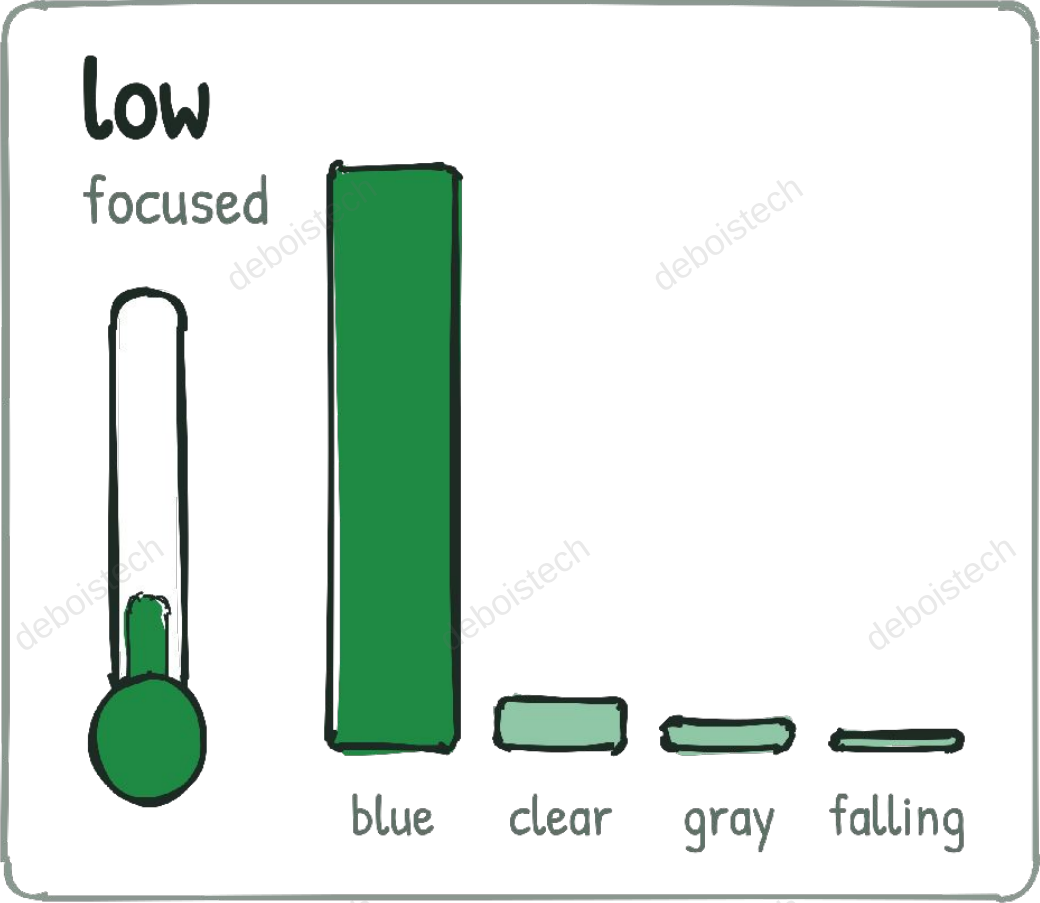
Lost in the middle

Attention isn't even inside the window.



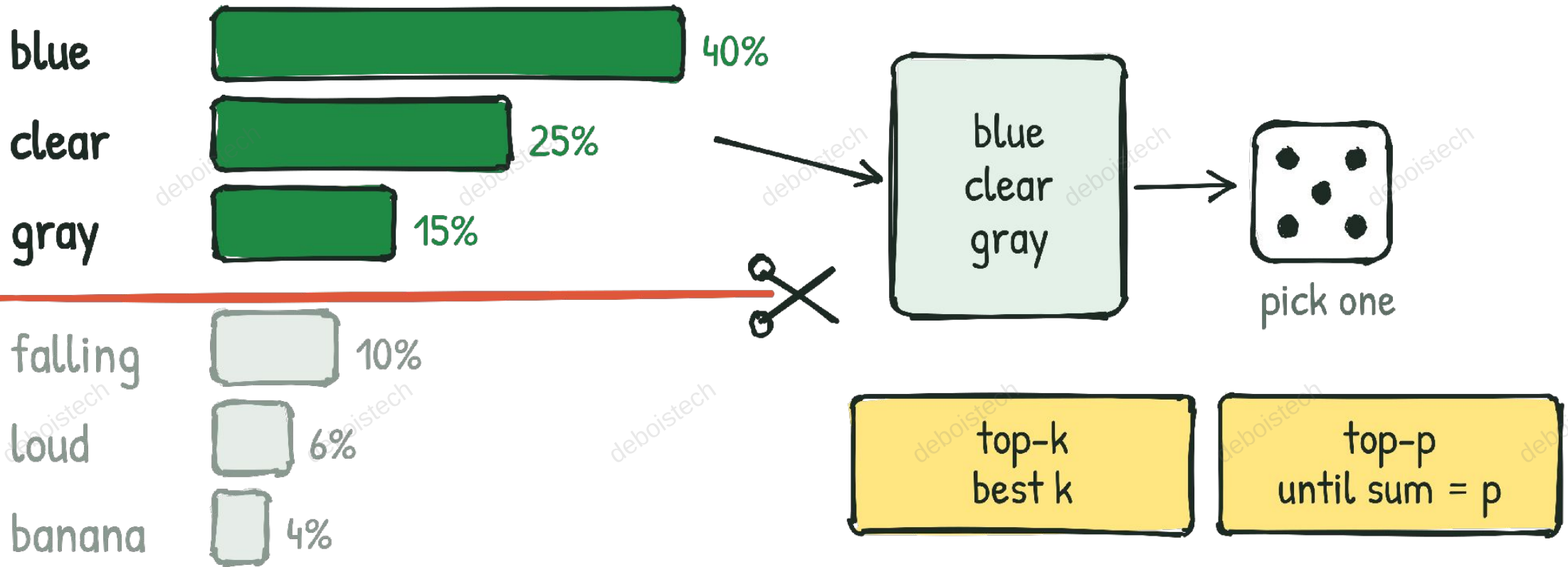
Temperature

Same prompt, different randomness.



Top-p · Top-k

Which tokens are even allowed next.



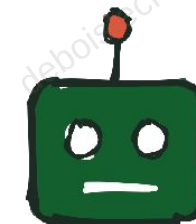
Hallucination

Confident, plausible, wrong.



Cite a paper on token drift.

Whitfield et al., Nature, 2017.



NOT REAL

✓ confident

✓ plausible

✗ wrong

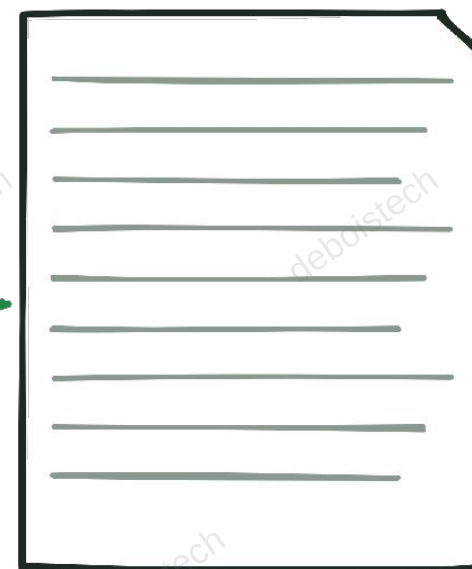
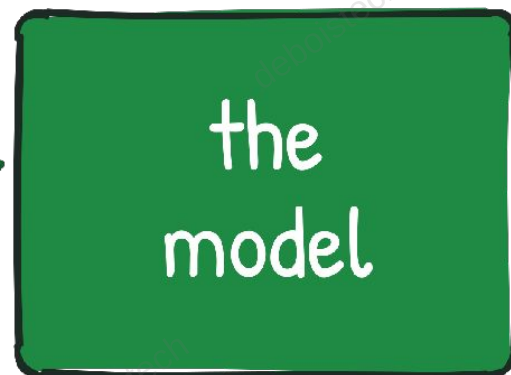
Prompt vs. training data

What it knows vs. what you told it.



baked in
static

training data

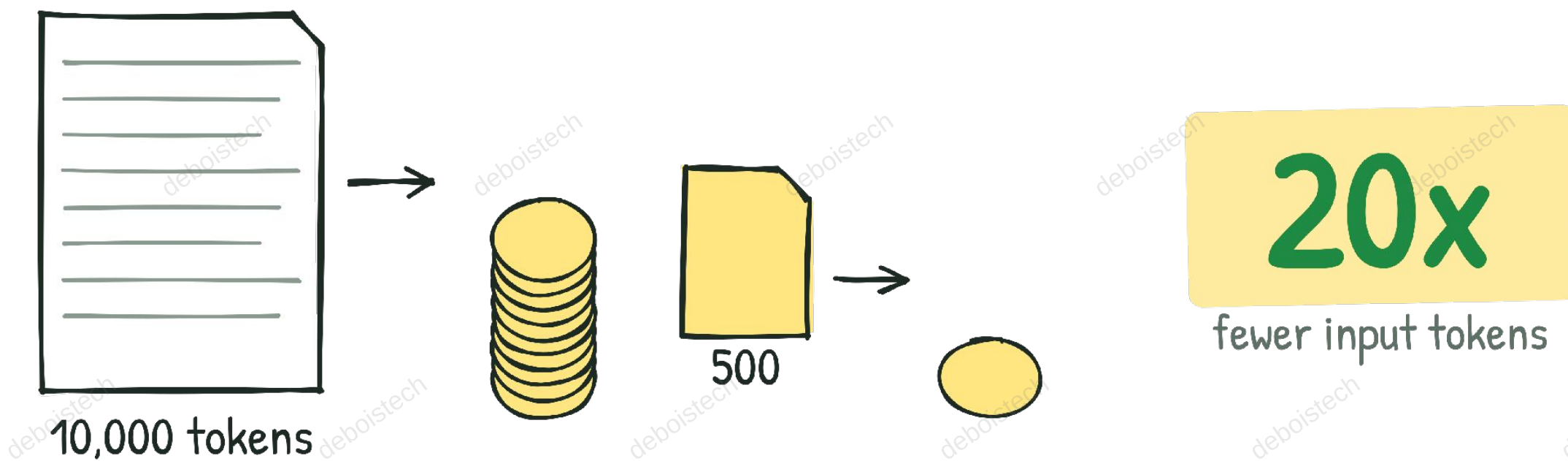


per call
yours
context

knows $\neq$ told

Token economics

Billed per token. Input and output differ.



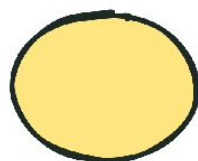
   priced differently

Input · output · cached

What each kind of token costs.

input

what you send:
prompt, history, docs



1x

output

what the model writes,
one token at a time



~4x

cached

repeated prefix,
re-used not re-read



a fraction

$$\text{cost} = (\text{input} \times \text{price}) + (\text{output} \times \text{price})$$

illustrative ratios

Prompt caching

Repeated prefixes get cheaper.

call 1

System

Long context

New
message



call 2

System
cached

Long context
cached

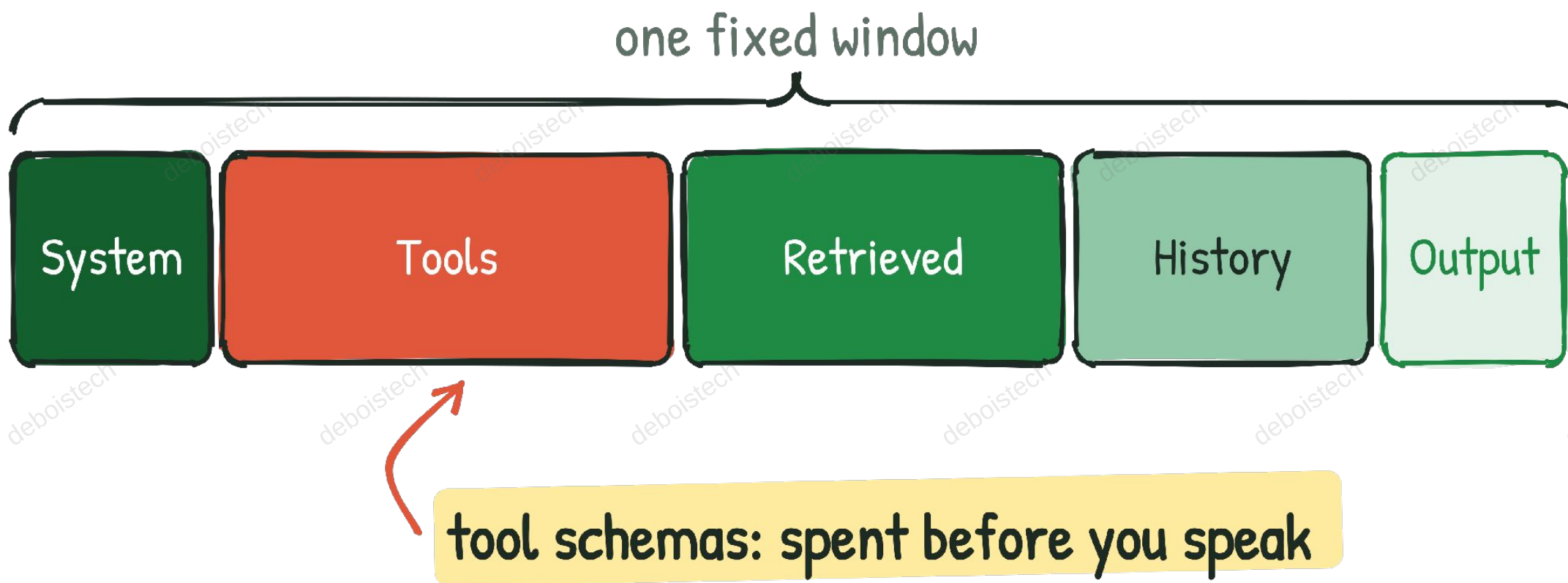
New
message



the 2nd call costs less

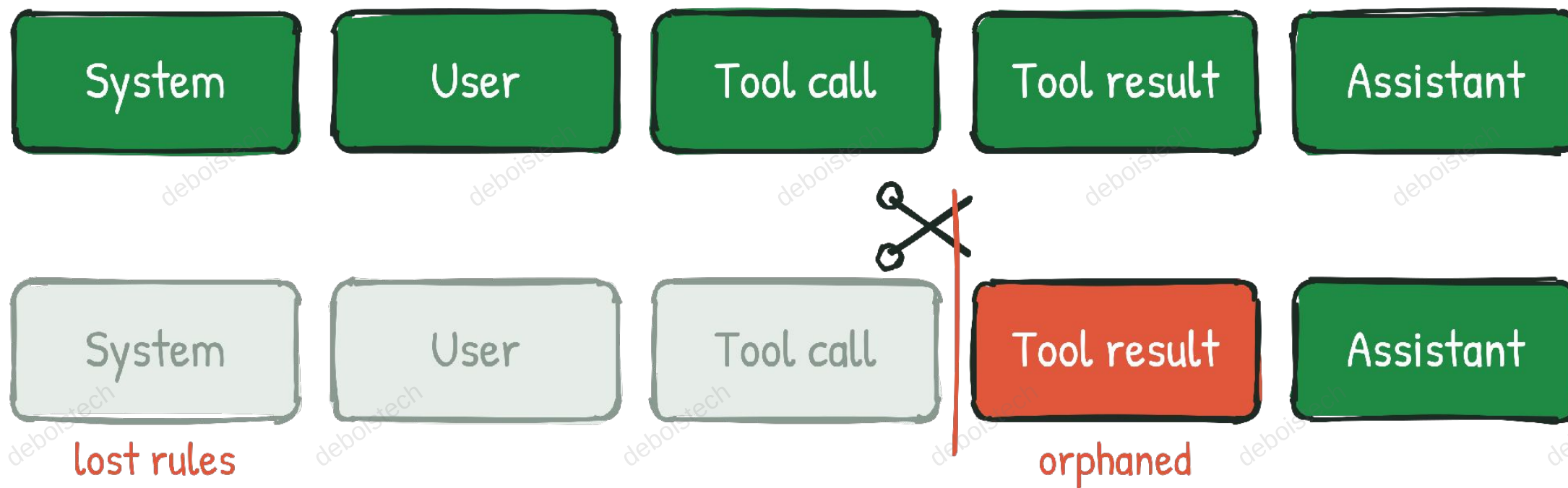
Token budgeting

Everything competes for one window.



Silent truncation bugs

Naive truncation drops the oldest first.



😞 no error. it just gets worse.

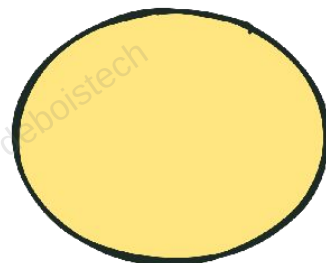
Prompts cost time, too

Not just correctness and cost.



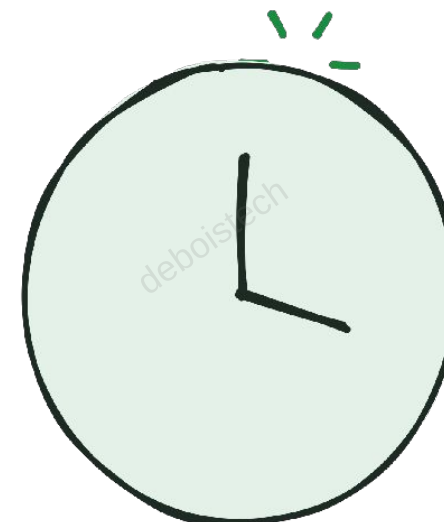
correct

+



cost

+



TIME

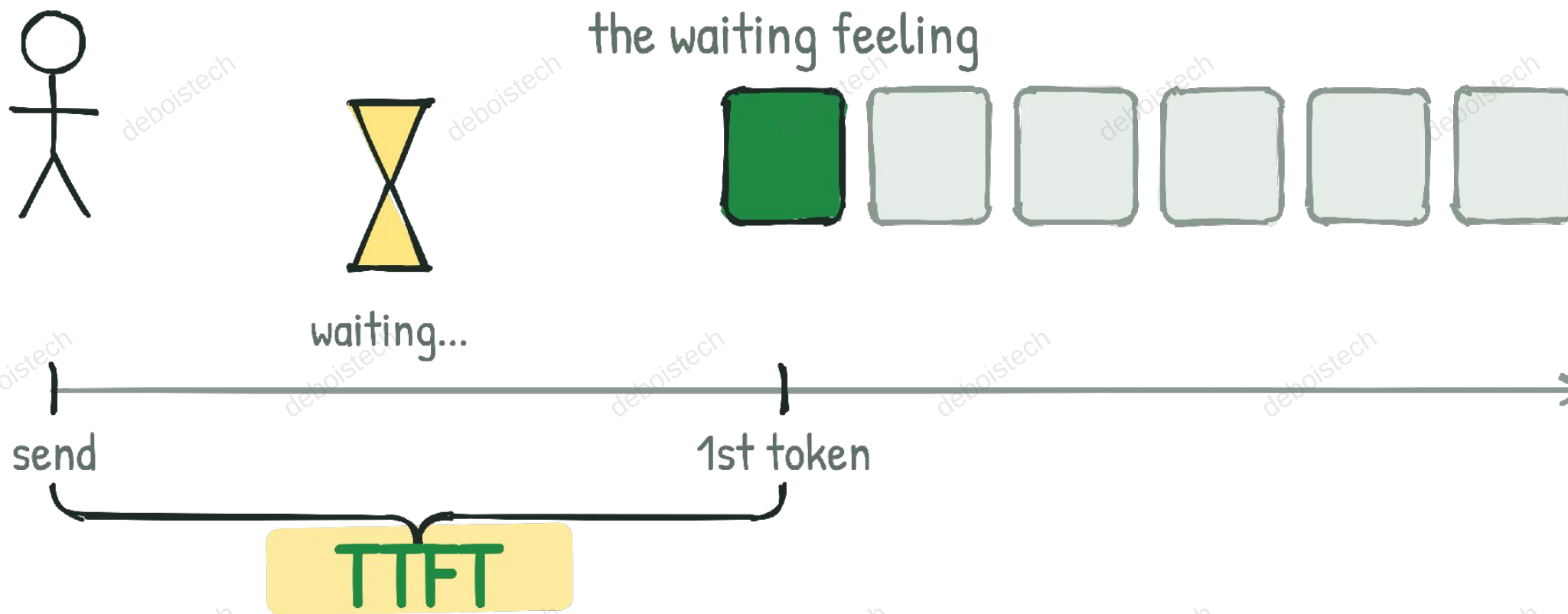
today: definitions



day 3: production calls

TTFT

Time to first token.



TPOT

Time per output token.



the typing-speed feeling



send

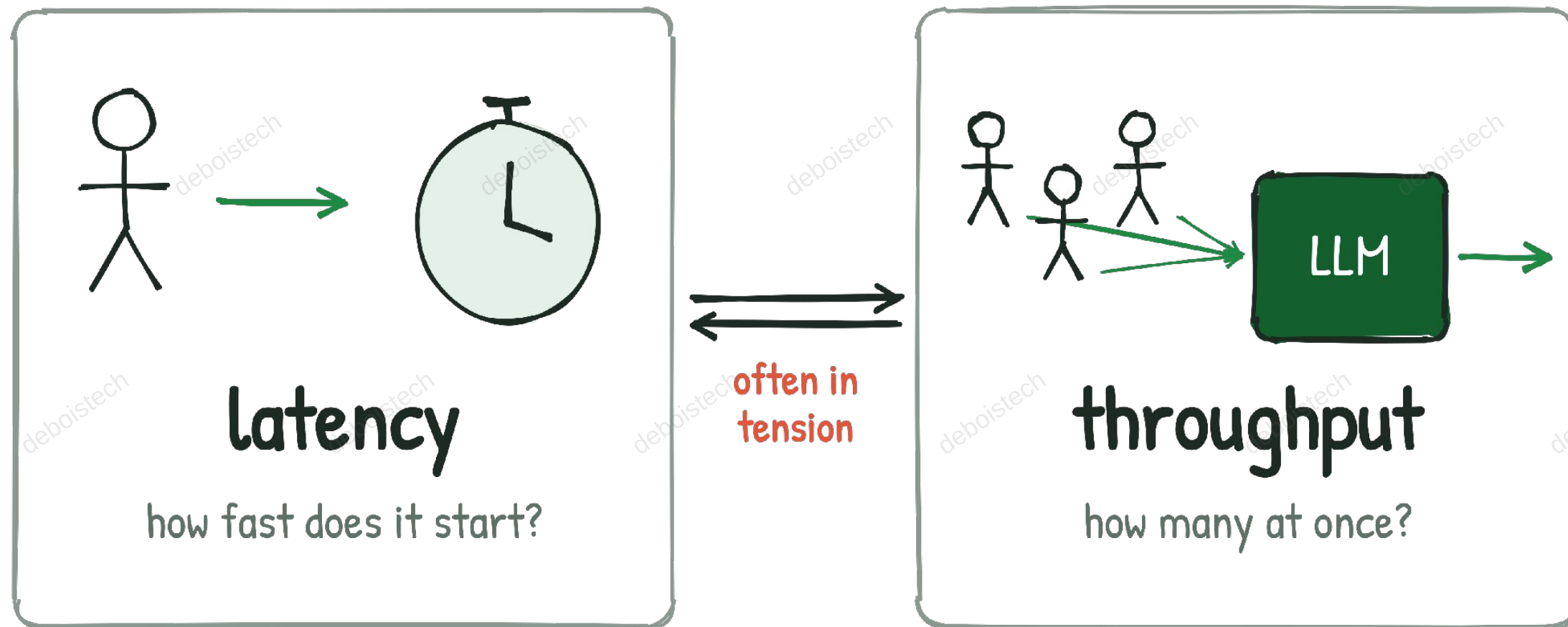
1st token



TPOT

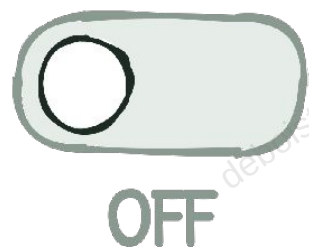
Latency vs. throughput

Start fast vs. serve many.

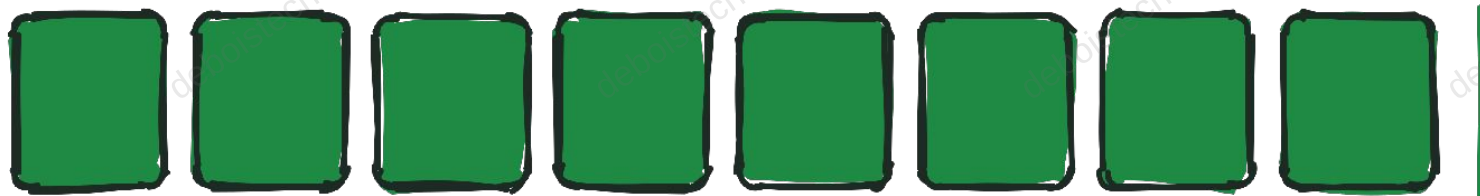


Streaming

Why text appears word by word.



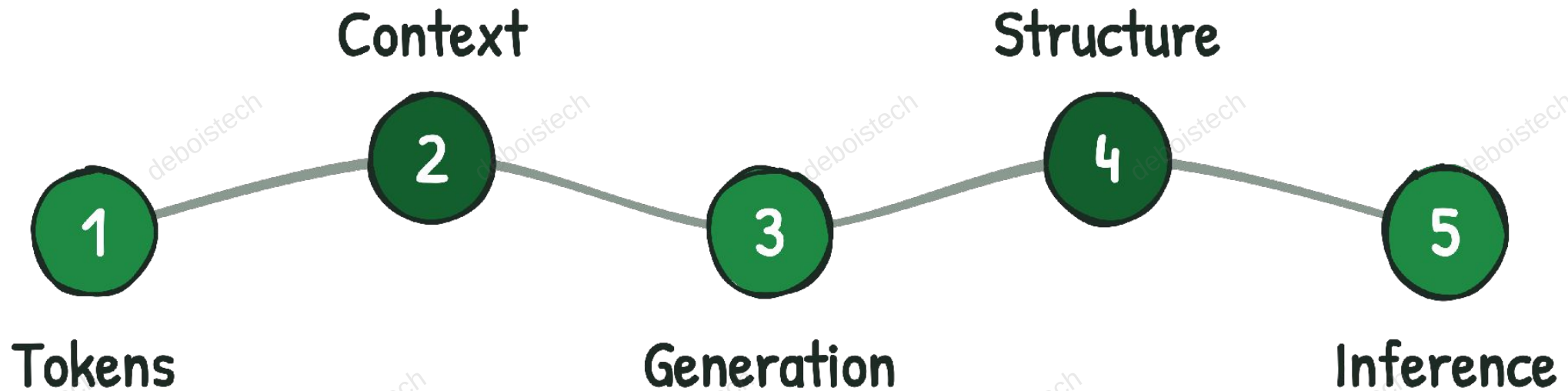
waiting...



same total time, different feel

Recap

Five ideas, one answer.



**Chats forget because tokens fall out
of a fixed window. A buffer limit, not memory loss.**

Quiz Time!

Four options, one answer.

kahoot.com

ABC

Up next: Day 2 & 3

Where we go from here.

